by the apps to indicate that the device should not be put to sleep. However, bugs in mobile apps can cause 'wakelocks' to be misused, resulting in the device continuing to drain energy even when not in use. For instance, if an application has acquired a 'wakelock' at some point in time, it should release the 'wakelock' to allow the device to go back to sleep. Failure to release the 'wakelock' can cause continuous drain of the battery. There has been considerable research work on detecting 'wakelock' bugs, also known as 'no-sleep' bugs. A detailed discussion of 'wakelock' bugs can be found in the paper *http://web1.cs.columbia.edu/~junfeng/reliable-software/papers/no-sleep.pdf*.

While the previous research papers we have discussed focus on detecting energy bugs in smartphones by analysing the individual device's behaviour, detecting energy consumption anomalies by analysing the behaviour in individual phones is difficult since the energy consumption can vary widely due to different device configurations, different apps co-running simultaneously, and usage under different environmental conditions. A novel direction in pinpointing energy bugs and high energy consumptions has been proposed in the free mobile app, 'Carat', which is available from *http://carat.cs.berkeley.edu/*.

Carat uses a collaborative approach to detect anomalies by collecting energy information from a wide variety of devices and uses that information to detect anomalous behaviour on an individual smartphone. Each smartphone on which Carat is installed, periodically sends information on battery usage, configuration settings, the set of applications installed and set of apps running currently to a centralised server. Data from a wide number of devices is collected, stored and analysed in the centralised database, to infer statistically, a normal energy consumption pattern for a device. If the measurements reported from the device deviate considerably from the expected consumption, an energy consumption anomaly is flagged by the app.

While the tools we have discussed so far help to detect abnormal energy consumption by applications in smartphones, as an app developer, you would be interested in making sure that your app is 'well-behaved' with respect to energy consumption. The open source tool JouleUnit is an energy profiling and testing framework, and is available at *https://code.google.com/p/jouleunit/*. The JouleUnit workbench is integrated into the Eclipse IDE, and allows you to construct and trigger energy tests for your app from the Eclipse IDE.

Learning to develop Android apps

One of our student readers, Ankith Subramanya, had written to me, sharing his experience on how he rapidly learnt Android app development. Given the enormous interest in mobile app development even from non-core programmers, I felt that it would be useful to share his experience in this column, for the benefit of those who are interested in learning Android app development. Here are Ankith's comments: